

5 Viruses can infect bacteria.

(a) Which virus can infect bacteria?

(1)

- ☐ **A** Ebola virus
- ☐ **B** human immunodeficiency virus (HIV)
- ☐ **C** lambda phage (λ phage)
- ☐ **D** tobacco mosaic virus (TMV)

(b) Some viruses that infect bacteria cause the production of molecules called holins.

Holins form protein channels in the cell membranes of bacteria. This allows polar molecules called lysins to reach the cell wall by facilitated diffusion.

The DNA of these viruses codes for lysins.

(i) Describe the role of channel proteins in the facilitated diffusion of lysins.

(2)

(ii) Explain how the primary structure and the tertiary structure of holins determine the properties of these channel proteins.

(3)

(iii) Lysins break down the cell wall of bacteria.

Explain the role of lysins in the lytic cycle.

(2)

(Total for Question 5 = 8 marks)